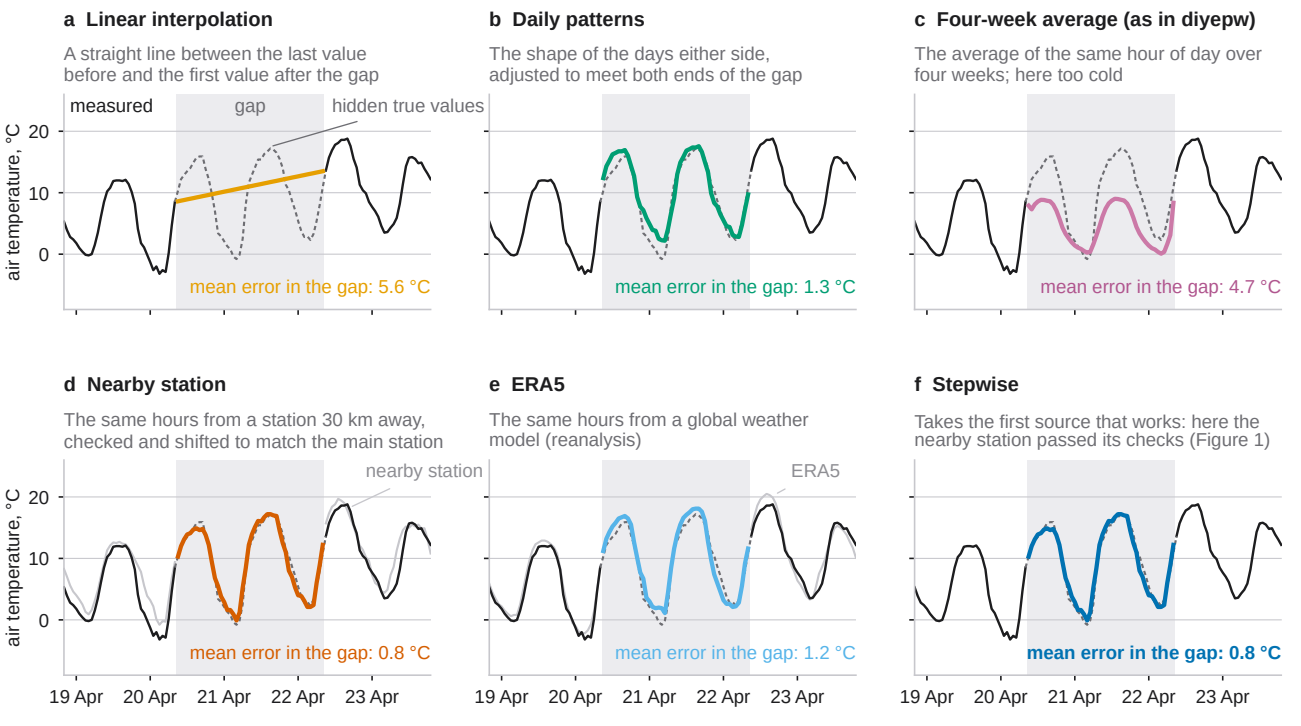
a Linear interpolation
A straight line between the last value before and the first value after the gap
measured
gap
hidden true values
air temperature, °C
20
10
0
mean error in the gap: 5.6 °C
b Daily patterns
The shape of the days either side, adjusted to meet both ends of the gap
mean error in the gap: 1.3 °C
c Four-week average (as in diyepw)
The average of the same hour of day over four weeks; here too cold
mean error in the gap: 4.7 °C
d Nearby station
The same hours from a station 30 km away, checked and shifted to match the main station
nearby station
mean error in the gap: 0.8 °C
e ERA5
The same hours from a global weather model (reanalysis)
ERA5
mean error in the gap: 1.2 °C
f Stepwise
Takes the first source that works: here the nearby station passed its checks (Figure 1)
mean error in the gap: 0.8 °C
19 Apr
20 Apr
21 Apr
22 Apr
23 Apr